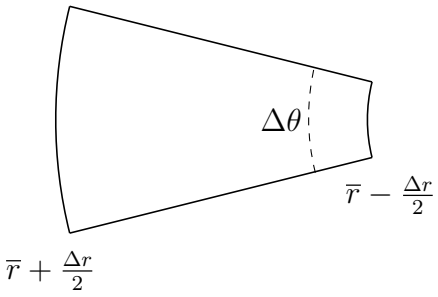
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Double Integrals in Polar Coordinates

Consider the iterated integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2)\, dy\, dx$. As an exercise left for the viewer, attempt to evaluate this iterated integral as written. It turns out that it is much more easily done in polar coordinates. But first we need some theory!

A polar "sub-rectangle" can be given by the bounds, $\overline{r} - \frac{\Delta r}{2} \leq r \leq \overline{r} + \frac{\Delta r}{2}$ and $\alpha \leq \theta \leq \alpha + \Delta\theta$. To compute the area of these "sub-rectangles" we use the formula for the area of a sector, which is equal to the fraction $\frac{\Delta\theta}{2\pi}$ of the area of the circle πr^2: $A = \frac{\Delta\theta}{2\pi}(\pi r^2) = \frac{r^2 \Delta\theta}{2}$.

Let's determine that area ΔA bounded by $\overline{r} - \frac{\Delta r}{2} \leq r \leq \overline{r} + \frac{\Delta r}{2}$ with a central angle of $\Delta\theta$:



$$\Delta A = \frac{(\overline{r} + \frac{\Delta r}{2})^2 \Delta\theta}{2} - \frac{(\overline{r} - \frac{\Delta r}{2})^2 \Delta\theta}{2} = \overline{r} \Delta r \Delta\theta.$$

As $\Delta r \to 0$ and $\Delta\theta \to 0$, $\overline{r} \to r$. Thus in polar coordinates, $\boxed{dA = r\, dr\, d\theta}$.

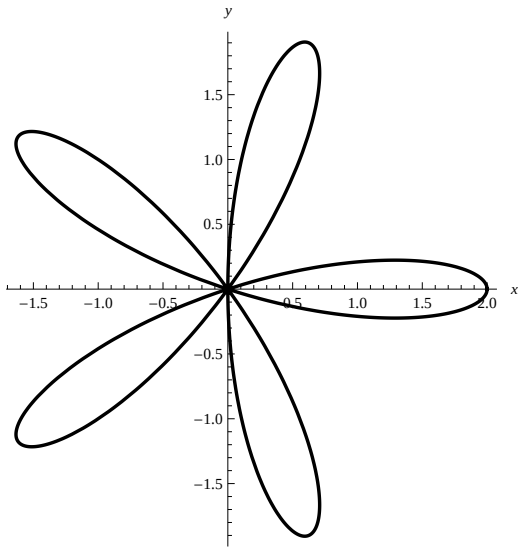
Of course $f(x, y)$ needs to be converted to $f(r, \theta)$ by using $x = r\cos(\theta)$ and $y = r\sin(\theta)$.

So $\int_0^1 \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2)\, dy\, dx = \int_0^{\frac{\pi}{2}} \int_0^1 (1 - r^2)\, r\, dr\, d\theta = \frac{\pi}{2}(1/2 - 1/4) = \frac{\pi}{8}$.

If a polar region R is given by $g(\theta) \le r \le h(\theta)$ and $\alpha \le \theta \le \beta$ then

$$\int\int_R f(r,\theta)dA = \int_\alpha^\beta \int_{g(\theta)}^{h(\theta)} f(r,\theta)\, r\, dr\, d\theta.$$

Let's find the area of the region bounded by one "leaf" of the rose $r = 2\cos(5\theta)$:



Notice that $\cos(5\theta)$ is an even function. We find the first time after $\theta = 0$ that the function equals 0. That occurs at $\theta = \frac{\pi}{10}$. Hence the area is (using symmetry)
$2\int_0^{\frac{\pi}{10}} \int_0^{2\cos(5\theta)} r\, dr\, d\theta = 2\int_0^{\frac{\pi}{10}} \frac{1}{2}(2\cos(5\theta))^2 d\theta = 4\int_0^{\frac{\pi}{10}} \frac{1}{2}(1+\cos(10\theta))d\theta = \frac{\pi}{5}.$

Let's evaluate the integral of $\displaystyle\iint_R \frac{1}{\sqrt{x^2+y^2}}\, dA$ where R is the region given by
$1 \le x \le \sqrt{4-y^2}$.

The region is to the right of $x = 1$, which in polar coordinates is $r = \sec(\theta)$, and to the left of $x^2 + y^2 = 4$, which in polar coordinates is $r = 2$. These curves intersect when $\sec(\theta) = 2$, which occurs at $\theta = \pm\frac{\pi}{3}$.

So the integral is $\displaystyle\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \int_{\sec(\theta)}^{2} \frac{1}{r}\, r\, dr\, d\theta = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 2 - \sec(\theta)\, d\theta = 2\left(\frac{2\pi}{3} - \ln(2+\sqrt{3})\right)$, using even-function symmetry.

The graph of $f(x) = e^{-x^2}$ has the classic "bell-shape" to it. Suppose we would like to know the area under this bell-shaped curve on $(-\infty, \infty)$. That is, $\int_{-\infty}^{\infty} e^{-x^2} dx$.

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \lim_{t\to\infty} \int_{-t}^{t} e^{-x^2} dx = \lim_{t\to\infty} \sqrt{\int_{-t}^{t} e^{-x^2} dx \int_{-t}^{t} e^{-y^2} dy} =$$

$$= \lim_{t\to\infty} \sqrt{\int_{-t}^{t}\int_{-t}^{t} e^{-x^2-y^2} dxdy} = \lim_{t\to\infty} \sqrt{\int_{0}^{2\pi}\int_{0}^{t} e^{-r^2} r dr d\theta} =$$

$$= \lim_{t\to\infty} \sqrt{\int_{0}^{2\pi} \left(-\frac{1}{2}\left(e^{-t^2} - 1\right)\right) d\theta} = \lim_{t\to\infty} \sqrt{-\pi\left(e^{-t^2} - 1\right)} = \sqrt{\pi}.$$